Checkpoint 1 Report

Does Genre Make an Indian Film Pay Off?

1. Research Objective & Hypothesis

Recent Indian film industry operations indicate to the assumption that high-budget action spectacles guarantee superior commercial success. This project tests that assumption by shifting the focus from **absolute box-office gross** to **capital efficiency**. Our primary objective is to determine if **action films** genuinely yield a higher **Revenue-to-Budget ratio** (we will call it ROI from now onwards) compared to other genres, or if lower-budget genres (e.g., Comedy, Drama) offer statistically superior or comparable financial multipliers.

2. Data Acquisition & Bias Mitigation

To accurately model financial returns, we must "try" to account for **survivorship bias**; the tendency of public databases to index only blockbuster hits while ignoring commercial flops. We aggregated **7 distinct open-source datasets** (including Kaggle box-office archives, IMDb metadata, and an IIMA case study dataset) spanning releases from **2005 to 2023**. By combining these sources, we again "tried to" capture a holistic financial spectrum, ranging from micro-budget failures to tentpole blockbusters.

3. Automated Engineering Pipeline

We engineered a modular, version-controlled Python pipeline pushed to our GitHub Repository The_ticket_ransom to transform messy, multi-source records into a reproducible, analysis-ready subset:

- **Ingestion & Standardization (`clean_data.py`):** We parsed disparate string formats, extracted normalized 4-digit release years via regular expressions, and standardized all financial metrics into uniform units of **Crores** (applying 10^7 scaling to raw INR datasets).
- **Hierarchical Conflict Resolution (`consolidate_data.py`):** Overlapping datasets frequently reported conflicting budgets. We grouped duplicates via a normalized alphanumeric title key and applied a **Domain Reliability Hierarchy**: *Modern Box Office* > *IIMA Case Studies* > *General Archives*.
- **Schema Projection & Strict Filtering (`generate_compact.py` & `filter_subset.py`):** We condensed the dataset into a strict 7-column schema and purged rows missing critical variables. The core target metric was explicitly computed as: **ROI Ratio = India Net Revenue / Budget**.